

Auto-Director AI — User Guide & Architecture

A Complete Guide to Automated 9:16 Micro-Drama Video Generation

1. What is Auto-Director?

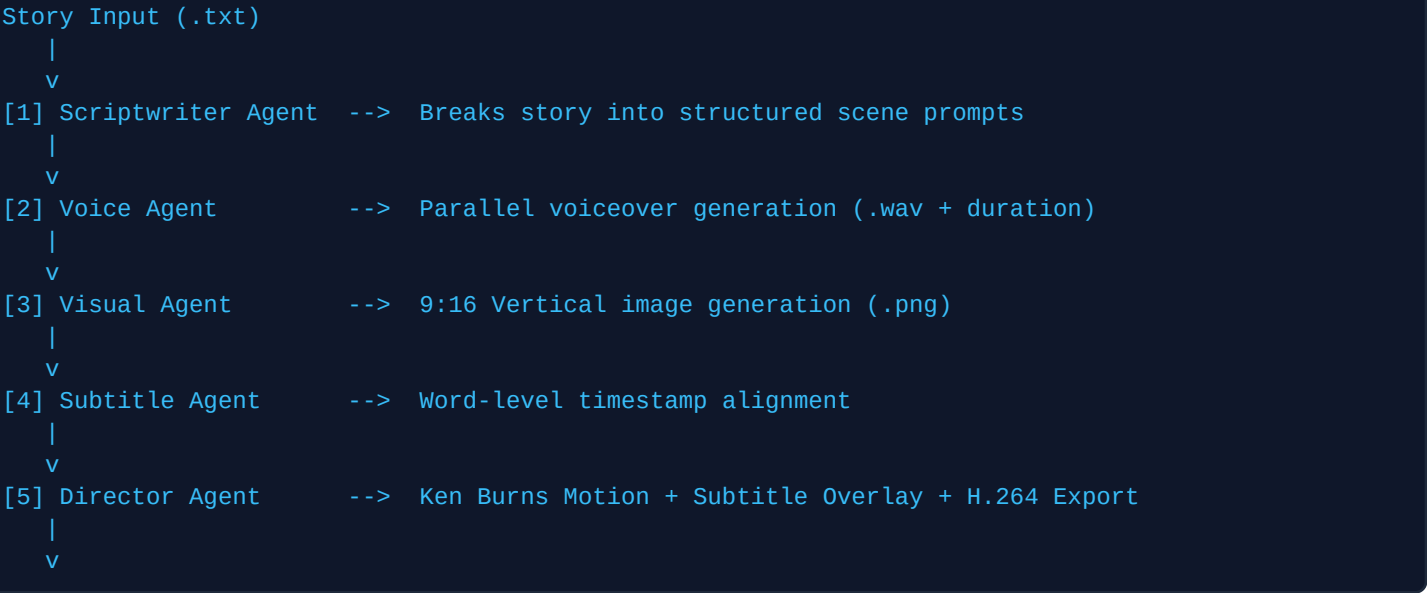
Auto-Director is an autonomous multi-agent AI system that converts raw text stories (in English or Hindi) into fully animated, voice-narrated, subtitle-aligned **9:16 vertical micro-drama videos** (ready for YouTube Shorts, Instagram Reels, and TikTok).

It replaces manual video editing by orchestrating 5 specialized AI agents working together in a unified pipeline.

2. How the 5 AI Agents Work (Architecture)

1. Scriptwriter Agent (The Writer) Analyzes the text story and breaks it into scenes, defining narration text, visual prompt details, and emotional mood.
2. Voice Agent (The Voice Actor) Synthesizes parallel audio narration using ElevenLabs or gTTS and records exact millisecond audio durations.
3. Visual Agent (The Artist) Generates high-resolution 9:16 vertical imagery using Gemini Flash 3.1, Pollinations Turbo, or Stock Photos with local caching.
4. Subtitle Agent (The Synchronizer) Calculates precise word-level start and end timestamps so subtitles appear in perfect audio sync.
5. Director Agent (The Video Editor) Combines audio/images, applies Ken Burns pan/zoom motion, overlays yellow subtitles, and exports the final MP4 video.

3. Pipeline Flow & Data Lifecycle



Output Video: output/final_video.mp4 (1080x1920 @ 30 FPS)

4. Multi-Tier Fallback Architecture (Zero-Downtime)

If external APIs hit rate limits or payment quotas, the pipeline gracefully falls back without crashing:

Component	Primary (Live)	Secondary Fallback	Offline Fallback
Voice Speech	ElevenLabs AI Voice	Google TTS (gTTS)	espeak-ng
Scene Visuals	Gemini 3.1 Flash Image	Pollinations Turbo AI	Stock Photos / PIL Cards
Subtitles	Dynamic Devanagari (Hindi)	Latin (English)	DejaVuSans-Bold

5. Step-by-Step User Commands

Step 1: Install Linux Prerequisites

```
sudo apt-get update && sudo apt-get install -y ffmpeg espeak-ng
```

Step 2: Setup Python Virtual Environment

```
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

Step 3: Run Video Generation

Run in Offline Demo Mode (Free, no API keys needed):

```
python3 main.py sample_input/my_story_english.txt -o my_video.mp4
```

Run in Live AI Mode (Real Voices & Images):

```
export GEMINI_API_KEY="your_key"
export ELEVENLABS_API_KEY="your_key"
python3 main.py sample_input/my_story_english.txt -o my_video.mp4
```